Solution exercise set 7

Rizzo, exercise 8.4:

See the R script file. It was given in the exercise text that the maximum likelihood estimator (MLE) of the rate λ in the exponential distribution is $\hat{\lambda} = n/\sum_{i=1}^{n} X_i$. This can be derived by the following calculation:

$$
\begin{aligned}
L(\lambda) &= \prod_{i=1}^{n} f(x_i) = \prod_{i=1}^{n} (\lambda e^{-\lambda x_i}) = \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i} \\
l(\lambda) &= \ln L(\lambda) = \ln(\lambda)^n + \ln(e^{-\lambda \sum_{i=1}^{n} x_i}) = n \ln(\lambda) - \lambda \sum_{i=1}^{n} x_i \\
\frac{dl(\lambda)}{d\lambda} &= \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0 \qquad \Rightarrow \lambda = \frac{n}{\sum_{i=1}^{n} x_i}
\end{aligned}
$$

I.e. the MLE is given as: $\hat{\lambda} = \underline{\underline{\frac{n}{\sum_{i=1}^{n} X_i}}}$.

The estimate is $\hat{\lambda} = n/\sum_{i=1}^{n} x_i = 12/1297 = 0.0093$.

The bootstrap procedure give an estimated standard error,

$$\widehat{\mathrm{SD}(\hat{\lambda}^*)} = 0.0043$$

and estimated bias of

$$\widehat{\mathrm{bias}}(\hat{\lambda}) = \overline{\hat{\lambda}^*} - \hat{\lambda} = 0.0014$$

These numbers will of course vary a bit from simulation to simulation.

Rizzo, exercise 8.5:

The intervals obtained from a run with $B = 2000$ bootstrap repetitions are reported below.

interval type	lower limit	upper limit
standard normal	-0.0007	0.0166
basic	-0.0025	0.0132
percentile	0.0053	0.0211
BCa	0.0045	0.0181

We observe that the four methods give quite different results. The two first intervals actually have lower limits below 0, which since $\lambda > 0$ is not desirable. The reason is probably that the distribution of the estimator $\hat{\lambda}$ is neither normal nor symmetric. The percentile method can by the way it is constructed not give negative values in this case (since $\hat{\lambda}^*$ cannot get negative). The most advanced BCa interval is a bit more narrow than the percentile interval.

Rizzo, exercise 11.3:

All in R

Rizzo, exercise 11.6:

All in R

Problem 1:

All in R

Problem 2:

In both cases, it suffices to show that $q(\theta_{t-1}|\theta^*)/q(\theta^*|\theta_{t-1}) = 1$: Normal random walk proposal:

$$\frac{q(\theta_{t-1}|\theta^*)}{q(\theta^*|\theta_{t-1})} = \frac{\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\theta_{t-1}-\theta^*)^2}{2\sigma^2}\right)}{\frac{1}{\sqrt{2\pi\sigma^2}} \left(-\frac{(\theta^*-\theta_{t-1})^2}{2\sigma^2}\right)} = \frac{\exp\left(-\frac{(\theta_{t-1}-\theta^*)^2}{2\sigma^2}\right)}{\exp\left(-\frac{(\theta^*-\theta_{t-1})^2}{2\sigma^2}\right)}$$

Now since $(\theta_{t-1} - \theta^*)^2 = (\theta^* - \theta_{t-1})^2$, it is clear that $q(\theta_{t-1}|\theta^*)/q(\theta^*|\theta_{t-1}) = 1$ in this case.

Uniform random walk proposal: write $q(a|b)$ as [1] $(2w)^{-1}\mathbf{1}_{[b-w,b+w]}(a)$.

$$\frac{q(\theta_{t-1}|\theta^*)}{q(\theta^*|\theta_{t-1})} = \frac{(2w)^{-1}\mathbf{1}_{[\theta^*-w,\theta^*+w]}(\theta_{t-1})}{(2w)^{-1}\mathbf{1}_{[\theta_{t-1}-w,\theta_{t-1}+w]}(\theta^*)} = \frac{\mathbf{1}_{[\theta^*-w,\theta^*+w]}(\theta_{t-1})}{\mathbf{1}_{[\theta_{t-1}-w,\theta_{t-1}+w]}(\theta^*)}$$

Clearly, the indicator function in the denominator will always evaluate to 1 (as θ^* will always be simulated where the proposal has positive probability density). The indicator function in the numerator will also always evaluate to 1 as the distance between the proposal and the old state is always smaller than w. Hence, we can also do RWMH with this proposal distribution.

[1] $\mathbf{1}_A(x) = 1$ if $x \in A$ and $\mathbf{1}_A(x) = 0$ if $x \notin A$.